

ECE8803-AI4

Module: Defect Classification

Lecture 2: Wafer Maps — Physics Beats Parameters, Until It Doesn't

Course website: <https://aikhan123.github.io/khan-lab-website/ai-semiconductors-course.html>

YouTube playlist: <https://www.youtube.com/watch?v=JOOhPoMm58I&list=PLF-ca21MPQt4>

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RECAP

Where Part 1 left us

*Every model encodes an assumption about the physics. **Its failures tell you which assumption broke.***

0.777

CNN macro F1 — our best

14 ×

Scratch precision jump, LR → CNN

spatial

structure was the missing ingredient

- Logistic regression on flat pixels couldn't place a pattern; a deep network with a million weights barely moved the needle; the CNN's locality and translation invariance is what finally fixed it.
- **We ended on triumph — every class detected, Scratch caught nine times in ten. The story felt finished.**

THE REVERSAL

Now forget the network for a moment

*28 hand-built features. 18 seconds of compute.
Accuracy 0.952 — higher than your CNN.*

0.952

accuracy (polar + RF)

18 s

feature extraction on a laptop

0

GPUs used

No convolutions. No backprop. No GPU. Just physics turned into 28 numbers and a random forest.

HISTORY

A note on the timeline

- **The polar-feature approach is not the “new” idea. It came first.**
- Wu, Jang & Chen introduced hand-engineered radial/angular features and similarity ranking for WM-811K in 2015 — before deep learning dominated the field.
- We inverted the timeline deliberately: CNN first, then features, so the reversal lands.
- **The lesson is not “old beats new.” It is that a model shaped to the physics can be astonishingly strong — and fails in exactly the place the physics doesn’t hold.**

Wu, Jang & Chen (2015)

“Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets”

IEEE Trans. Semiconductor Manufacturing, 28(1), 2015.

doi:10.1109/TSM.2014.2364237

PART 2 · SECTION 1 · THE HAND-CRAFTED-FEATURE ERA

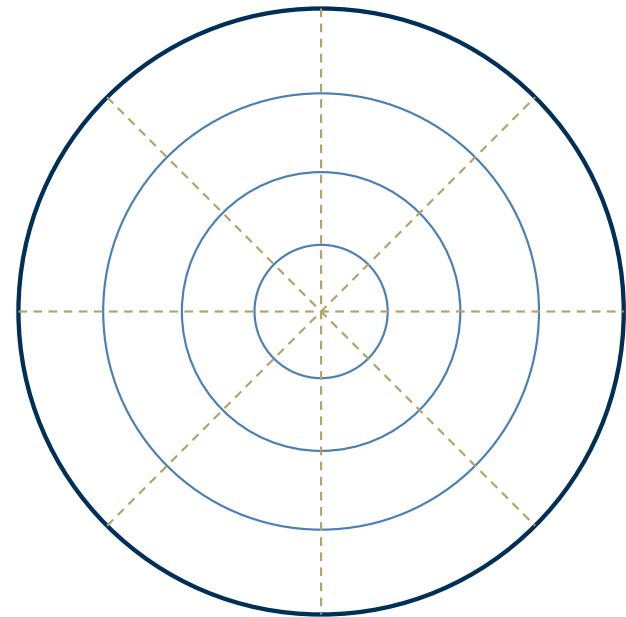
Physics as features

Encode the hypothesis directly: failure patterns are radially structured.

POLAR FEATURES

The idea: measure failures in polar coordinates

- Put the origin at the wafer center. Describe each map by where its failures fall in radius and angle.
- **Radial bins (annuli): fraction of dies failing in each concentric ring.**
- **Angular bins (wedges): fraction failing in each direction.**
- This bakes in the hypothesis: Center peaks at small r , Edge-Ring at large r , Donut in the middle.
- **Scratch would need the angular bins — a narrow directional spike. Hold that thought.**



radial annuli + angular wedges

POLAR FEATURES

28 features, in a few lines

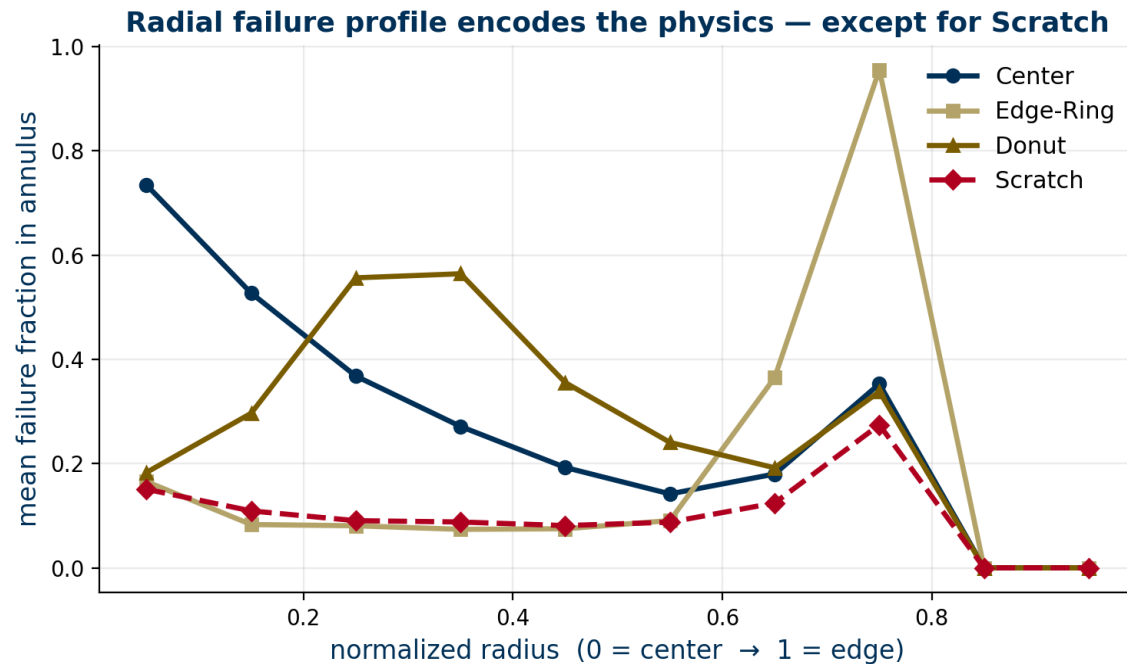
```
def polar_features(X):          # hypothesis: radially structured
    r = radius_of_each_die(H, W)    # 0 at center → 1 at edge
    th = angle_of_each_die(H, W)    # 0 ... 2π
    for each wafer m:
        fail = (m == 2)
        radial[b] = fail_fraction_in_annulus(b)    # 10 bins
        angular[b] = fail_fraction_in_wedge(b)    # 12 bins
        stats = [fail_frac, mean_radius,          # 6 summaries
                  radial_contrast, ...]
    return 10 + 12 + 6 = 28 features per wafer
```

The whole model

- **28 numbers per wafer.**
- Random forest, 300 trees.
- 18 s to extract all features.
- Fully interpretable — every feature has a physical meaning.

POLAR FEATURES

The hypothesis, visualized



- **Center peaks near $r = 0$.**
- **Donut peaks at mid-radius.**
- **Edge-Ring peaks near the edge.**
- Each radially-structured class has a distinct, separable signature — the features do real work.
- **Scratch is nearly flat: a thin line has no radial profile. The feature set is blind to it.**

RESULT

Polar features + random forest

0.952

accuracy (test)

0.712

macro F1 (test)

18 s

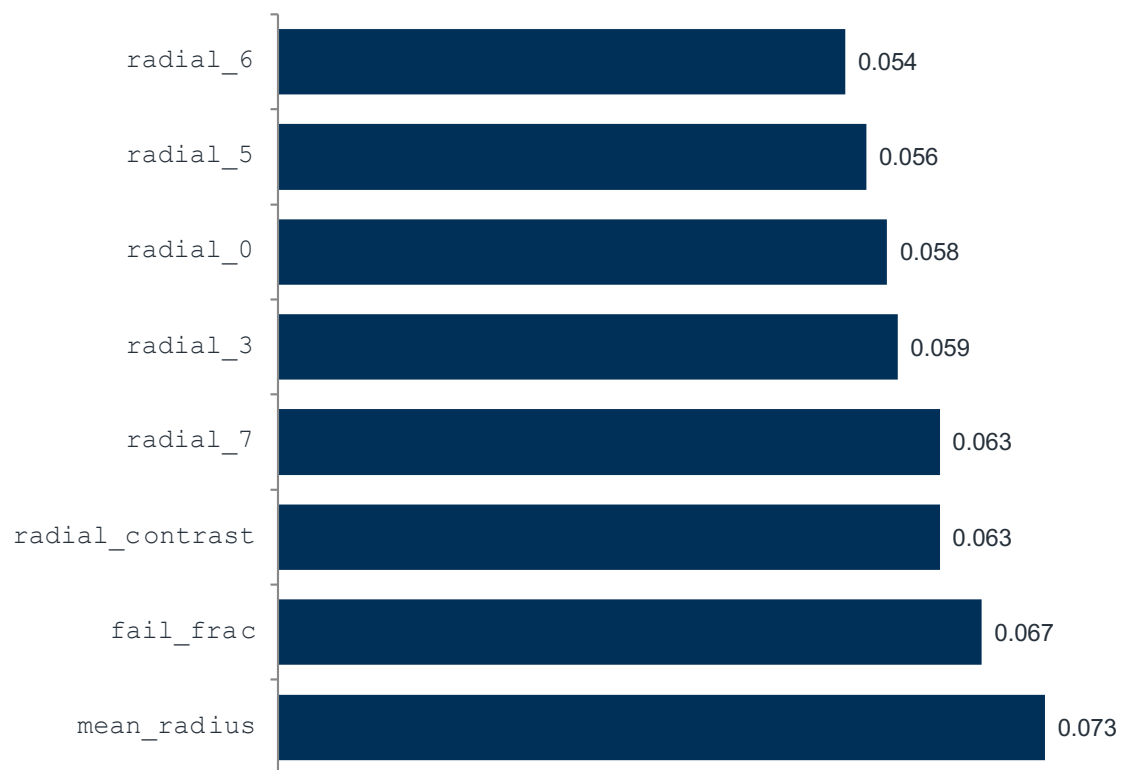
to extract features

Class	Prec	Recall	F1
none	0.958	0.995	0.977
Center	0.928	0.880	0.904
Edge-Ring	0.965	0.932	0.948
Near-full	0.909	1.000	0.952
Loc	0.737	0.355	0.480
Scratch	1.000	0.019	0.037

- Accuracy 0.952 — higher than the CNN's 0.939.
- On radially-structured classes it is excellent: Center and Near-full beat the CNN outright.
- But look at the Scratch row. Precision 1.000, recall 0.019.
- Something is very wrong — and very instructive.

POLAR FEATURES

What the forest actually used



- **Seven of the top eight features are radial.**
- The one exception, fail_frac, is a global count — also not angular.
- **The 12 angular features barely register in the top ranks.**
- The forest leaned entirely on the radial hypothesis — so anything without a radial signature is invisible.

THE REVEAL

Perfect, and useless

Scratch: precision 1.000 · recall 0.019

4 / 210

scratches found in the test set

100%

of those 4 were correct

89%

the CNN's Scratch recall, for contrast

The model is right every single time it says “Scratch” — and it almost never says it. **A detector that fires four times a year is not a detector.** The CNN, by contrast, caught 89% of scratches at 0.405 precision — noisier, but actually useful on the floor.

THE METRIC LESSON

Why F1 uses the harmonic mean

0.51

arithmetic mean of P and R

looks respectable

vs

0.037

F1 (harmonic mean)

tells the truth

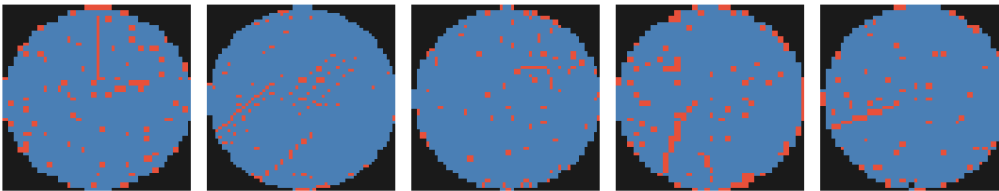
- $P = 1.000$, $R = 0.019$.
- Arithmetic mean forgives one number being near zero.
- **Harmonic mean is dominated by the smaller one — you cannot hide a broken recall behind perfect precision.**

$F1 = 2 \cdot P \cdot R / (P + R)$. With $R = 0.019$, $F1 \approx 0.037$ no matter how perfect the precision. This is the whole reason we report F1 instead of averaging — and why macro-F1, which averages per-class F1, punishes a model that quietly abandons a rare class.

DIAGNOSIS

The assumption that broke

Scratch — 1,193 wafers — thin line - physical contact, robot or chuck handling



- A scratch is a thin line at some orientation. Its failures are spread across many radii — no radial peak.
- **The information that could catch it lives in the angular features — a narrow directional spike.**
- But the forest learned that radial features carry the signal, so the angular bins got almost no weight.
- **The polar hypothesis — “patterns are radially symmetric” — is simply false for Scratch. The model’s failure names the broken assumption exactly.**

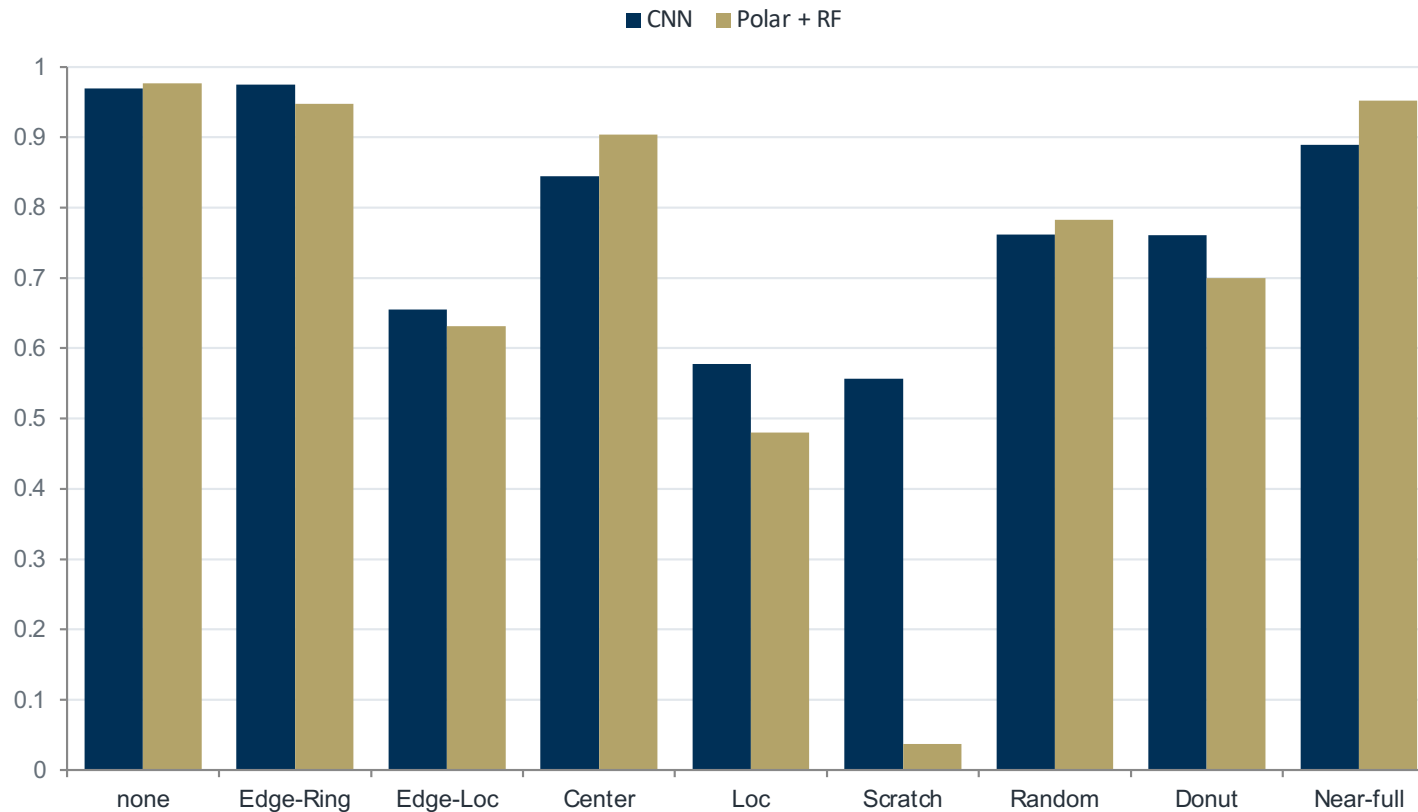
PART 2 · SECTION 2

Head-to-head

Two models with opposite inductive biases — and where each one breaks.

HEAD-TO-HEAD

CNN vs polar features, class by class



- They track closely on most classes.
- **Polar wins where physics is radial: Center, Near-full.**
- **One catastrophic gap: Scratch, 0.557 vs 0.037.**
- **Opposite biases, near-identical accuracy, one class that separates them.**

HEAD-TO-HEAD

Polar features beat the CNN — where physics holds

Class (F1)	CNN	Polar + RF	Winner
Center	0.845	0.904	Polar
Near-full	0.889	0.952	Polar
Loc	0.578	0.480	CNN
Scratch	0.557	0.037	CNN

Takeaway Polar+RF wins exactly the radially-structured classes it was designed for, and collapses only where the radial-symmetry hypothesis is false.

The features aren't "worse" than the CNN — they're a different bet on the physics, and the bet pays off precisely where the physics is right.

HEAD-TO-HEAD

The full scoreboard

Model	Accuracy	Macro F1	Scratch F1	Breaks on
Majority baseline	0.854	~0.10	0.000	everything
Logistic regression	0.688	0.473	0.053	variable position
Deep neural net †	0.818	0.437	0.048	position (same as LR)
CNN	0.939	0.777	0.557	ambiguous classes
Polar + RF	0.952	0.712	0.037	non-radial (Scratch)

Best accuracy and best macro-F1 belong to different models. There is no single winner — each model encodes a different assumption about the physics, and each one's failures point straight at the assumption that broke. **That is the whole course in one table.**

† The deep network (a million-weight fully-connected net) is on the 26K-wafer development subset; the other rows are the full test set. Even so it never beats logistic regression — depth was not the missing piece.

A footnote on the “bitter lesson”

Sutton, 2019

“The biggest lesson ... is that general methods that leverage computation are ultimately the most effective, by a large margin.”

The arc we drew in Part 1 — hand-crafted features giving way to learned representations — is exactly this lesson. **By that logic the CNN should crush 28 hand-built features.**

- **On accuracy, it doesn't — polar + RF ties it in 18 seconds, no GPU.**
- Why the exception holds here: labels are scarce (only ~25k named patterns) and the physics is strong and low-dimensional — so encoding it by hand is a great bet.
- **The nuance: the bitter lesson is about the limit of scale. Give the CNN more labeled data and it pulls ahead — it already wins Scratch, the class features can't see.**

Takeaway Match the method to the data regime: hand-crafted priors when labels are scarce; learned representations when data is abundant.

When both models fail on the same wafer

- The CNN and polar+RF have opposite inductive biases — pixels-with-locality vs radial-physics.
- So when they disagree, at least one is wrong. But when they agree and are both wrong, something deeper is going on.
- **Those “both-wrong” wafers are an empirical estimate of the label-noise floor: cases where the label itself is contestable.**
- **No architecture can fix a wafer whose true class two expert labelers would dispute.**

Why it matters

It tells you **when to stop tuning**. If your errors are converging on the both-wrong set, the remaining gap is label noise, not model capacity. Chasing it wastes effort — and risks fitting the noise.

PART 2 · SECTION 3

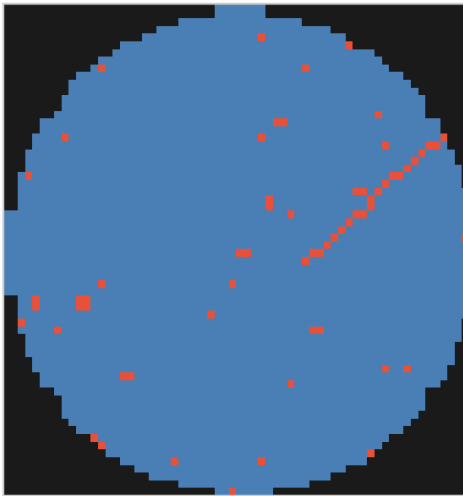
The preprocessing bug

A real error — invisible in aggregate, devastating for the one class you care about.

THE BUG

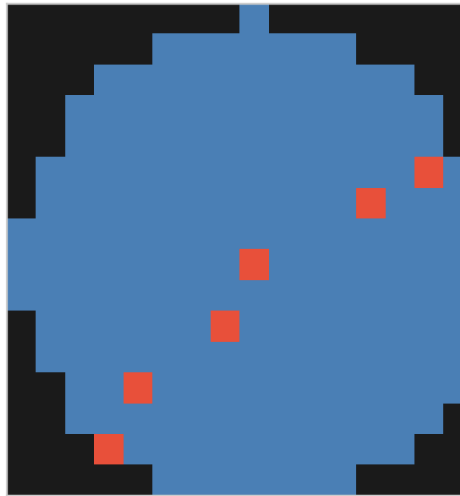
How a downsampling choice erased Scratch

Original 64×64



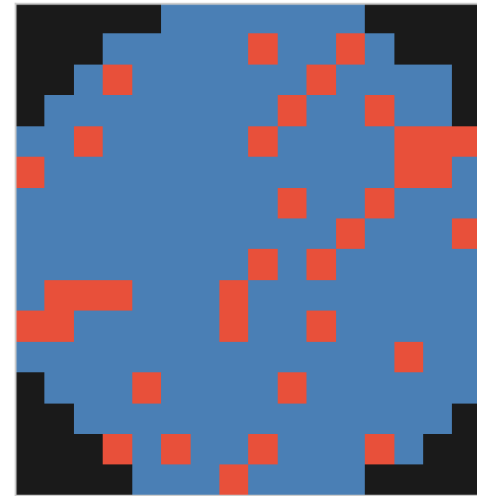
64 / 64 failing dies kept (100%)

Subsampled → 16×16



6 / 64 failing dies kept (9%)

Max-pooled → 16×16



36 / 64 failing dies kept (56%)

Takeaway Subsampling keeps every k-th row and column — a one-die-wide line falls through the gaps. Max-pooling marks a block failing if ANY die in it failed, so thin features survive.

THE BUG

Invisible in aggregate, fatal in the tail

```
# WRONG - subsampling: drops dies
ri = (arange(n)*h//n)
ci = (arange(n)*w//n)
out = m[ix_(ri, ci)]          # thin line lost

# RIGHT - max-pool: block fails if ANY die fails
out = maximum.reduceat(
    maximum.reduceat(m, rb, 0), cb, 1)
```

- Median wafer is 33×33; half the dataset is smaller than 32×32 — for most wafers, nothing was lost.
- **The damage was concentrated entirely in the large-grid minority.**
- **That minority is exactly where thin-feature classes like Scratch live.**
- So aggregate accuracy looked fine while the one class you care about quietly died.

Takeaway A bug that averages away to nothing can still destroy a rare class. Always audit preprocessing per-class, not just overall.

PART 2 · SECTION 4 · TOWARD THE FOUNDATION-MODEL ERA

The advanced landscape

Every technique ahead relaxes an assumption an earlier model made.

Closing the polar blind spot

- The diagnosis was precise: Scratch has no radial signature, and the angular bins carried no weight.
- **So add features that measure directionality directly:**
 - PCA elongation of the failing-die coordinates — a line has one dominant axis; a blob does not.
 - A Hough-style directional score — how well do the fails lie along a single line?
- **Add these and Scratch recall recovers, without touching the classes that already worked.**

The arc, completed

Students found the blind spot from the model's failures. Now they close it by adding exactly the feature the physics demanded — and measure the recovery. Diagnosis → fix → verification, the whole loop.

ADVANCED

Physically-valid augmentation

- Rotations and reflections are legitimate here — a scratch at 30° and one at 210° are the same failure mode.
- This is a domain argument, not a generic ML trick: the label is invariant to wafer orientation.
- It expands the rare classes (Scratch, Donut) for free, attacking the imbalance at its root.
- Caveat: if there is a systematic notch orientation, rotation destroys real information — check before you augment.

ADVANCED

Reshaping the loss, and inverting the problem

Focal loss

Down-weights easy examples so the 147k none wafers stop dominating the gradient. The network spends its capacity on the classes that are actually hard.

Anomaly detection

Train only on none; flag any wafer that deviates. Sidesteps imbalance entirely and catches novel excursion signatures with no label — realistic, since new fault modes appear in fabs constantly.

Takeaway Both relax the same assumption — that every class is worth equal gradient — from opposite directions.

Self-supervised pretraining

638k

unlabeled wafers, already prepared

- Only 21% of wafers are labeled — but the other 638,507 still contain the physics.
- **Pretrain a representation on all of them (masked reconstruction, contrastive views), then fine-tune on the small labeled set.**
- **This is the closest fit to the course thesis: learn structure from process data, then specialize with scarce labels.**
- The data is already split and ready in the homework bundle.

ADVANCED

Vision foundation models — with a caveat

- Few-shot classification with large pretrained vision models (NV-DINOv2, VLM zero-shot via NIM APIs) closes the loop to industry semiconductor-defect work.
- **But recall Part 1: wafer maps are not photographs. These models add little on categorical 0/1/2 grids.**
- They belong on real SEM / optical inspection images — where texture and appearance actually exist.
- Knowing which representation a method needs is the skill — matching the tool to the physics, not the hype.

ADVANCED

Three more the field actually uses

1

Active learning

Which wafers should we label next?
Spend the labeling budget where the model is most uncertain.

2

Similarity retrieval

What the 2015 paper did, and what yield engineers actually want: “show me wafers that look like this one.”

3

Deployment realities

Calibration drifts over time; an engineer who sees a 3%-precision alarm once too often stops trusting it. The sociology matters.

Takeaway Classification is the start of the pipeline, not the end — the value is in what you do with the prediction.

RECAP

One idea, five times — the whole unit

Model	Assumption it encodes	Where it breaks
Logistic regression	Pixels are independent features	position varies (Loc, Scratch)
Deep neural net	Depth and width, not spatial structure	position still varies (like LR)
CNN	Locality + translation invariance	precision on ambiguous classes
Polar + RF	Patterns are radially symmetric	a line has no radial signature
Advanced methods	Each relaxes a prior assumption	the moving frontier

Every model encodes an assumption about the physics. Its failures tell you which assumption broke. If you retain only this, the lectures worked — and it generalizes to the rest of the course.

ECE 8803-AI4

Questions

Dataset & citation: M.-J. Wu, J.-S. R. Jang, J.-L. Chen, “Wafer Map Failure Pattern Recognition and Similarity Ranking for Large-Scale Data Sets,” *IEEE Trans. Semiconductor Manufacturing*, 28(1), 1–12, 2015. doi:10.1109/TSM.2014.2364237

Hands-on exercise: WM-811K wafer maps · train / dev / test provided as .npz · akhan40@gatech.edu